3. Exercise

Data Science mit Python

Deadline: **06.01.26**, Midnight

Wintersemester 2025/26

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Agenda

- Questions?
- 3. Exercise

Question?

3. Exercise

- The exercise consists of **2 parts**

- 1) Bias and Fairness in Traffic Stops
  - Exploratory Data Analysis

- 2) Pre-Processing Data Sets (Vibe Coding)

Part 1: Bias and Fairness

Bias and Fairness in Traffic Stops

- This dataset contains records of traffic stops conducted by police officers across the United States from 2005 to 2015.
- The data was sourced from the Stanford Open Policing Project
  - A goal is to analyze and improve interactions between police and the public
- The dataset encompasses stops made across all U.S. states during the specified period.
- It has 85845 rows × 13 columns.

Dataset

- Look at the data

- The dataset contains the following columns:

```
'stop_date‘,     'stop_time‘,    'driver_gender',
'driver_age‘,    'driver_race‘, 'violation_raw',
'violation‘,     'search_conducted', 'search_type',
'stop_outcome', 'is_arrested', 'stop_duration‘,
'drugs_related_stop'
```

Dataset

- Familiarise yourself with the data set
- The data are stored in a flat CSV file in tabular form

stop_datetime	stop_date	stop_time	driver_gender	driver_age	driver_race	violation_raw	violation	search_conducted	search_type	stop_outcome	is_arre
2005-01-02 01:55:00	2005-01-02	01:55	M	20	White	Speeding	Speeding	False	NaN	Citation	
2005-01-18 08:15:00	2005-01-18	08:15	M	40	White	Speeding	Speeding	False	NaN	Citation	
2005-01-23 23:15:00	2005-01-23	23:15	M	33	White	Speeding	Speeding	False	NaN	Citation	
2005-02-20 17:15:00	2005-02-20	17:15	M	19	White	Call for Service	Other	False	NaN	Arrest Driver	
2005-03-14 10:00:00	2005-03-14	10:00	F	21	White	Speeding	Speeding	False	NaN	Citation	

Simple Types

```
'stop_date‘: String

'stop_time‘: String

'driver_age‘: Int64

'violation_raw‘: String

'search_conducted‘: Boolean

'is_arrested‘: Boolean

'drugs_related_stop‘: Boolean
```

Category Types

'driver_gender‘:Category:

['F', 'M‘]

'driver_race‘: Category:

['Asian', 'Black', 'Hispanic', 'Other', 'White‘]

'violation‘: Category:

['Speeding', 'Other', 'Equipment', 'Moving violation', 'Registration/plates', 'Seat belt']

'stop_outcome‘: Category

['Citation', 'Arrest Driver', 'N/D', 'Warning', 'Arrest Passenger', 'No Action']

'stop_duration‘: Category

['0-15 Min', '16-30 Min', '30+ Min']

List Types

`'search_type‘: List`

Values in:

nan, 'Incident to Arrest‘, 'Protective Frisk‘, 'Probable Cause‘, 'Reasonable Suspicion‘, 'Inventory'

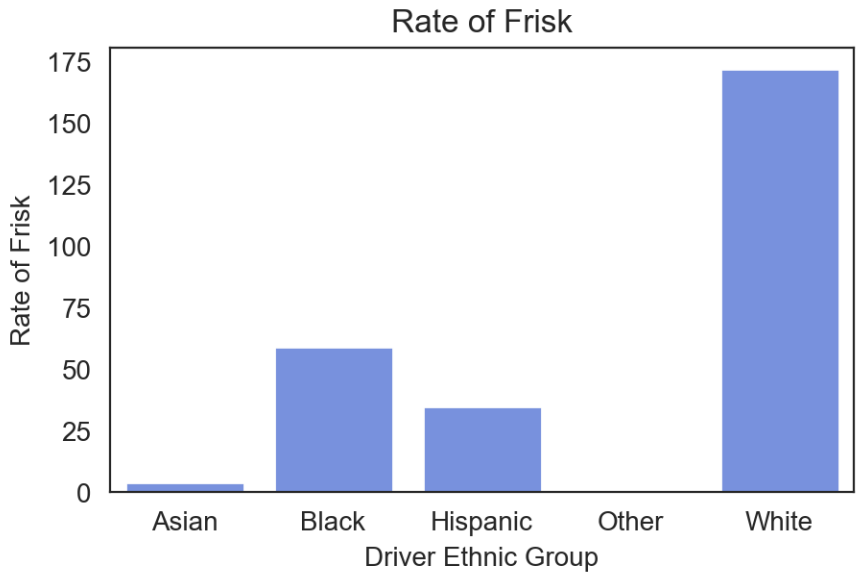
Key Questions

1. Assess the impact of **race, gender, and age** on the frequency of police traffic stops.
2. Evaluate how **gender influences** police behaviour during traffic stops.
3. Examine whether **gender** affects the likelihood of being **frisked** during a search.
4. Compare frisk rates during searches between **males and females**.
5. Compare frisk rates during searches between **Black and White drivers**.
6. Determine which **gender** is more likely to be arrested during a traffic stop.
7. Investigate the impact of **driver age** on the likelihood of being stopped.
8. Analyse whether **stop duration** influences the probability of an arrest.
9. Study how **stop duration** has changed over time to identify evolving policing practices.
10. Use a large language model to suggest additional insightful analyses of the dataset.

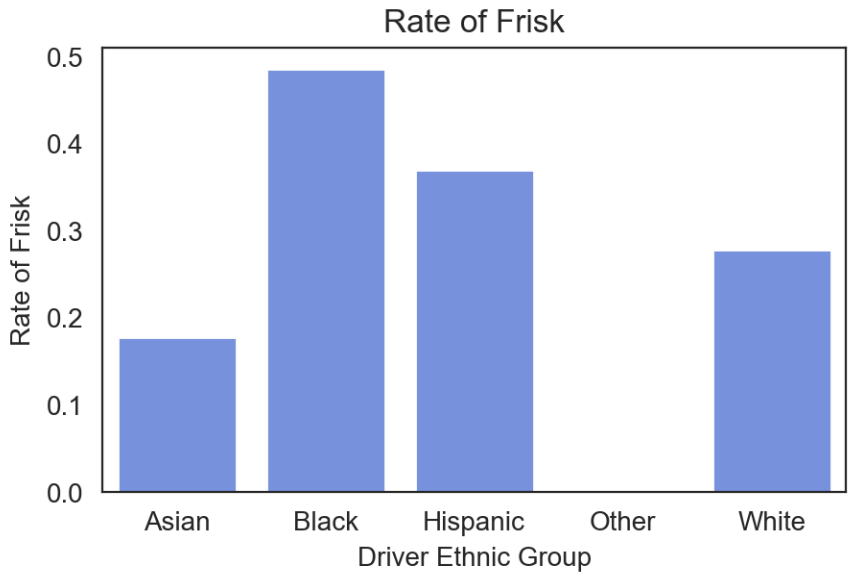
Relative vs Absolute Proportion

driver_race	count
White	61758
Black	12152
Hispanic	9448
Asian	2249
Other	238

Total number of stops



Absolute Numbers

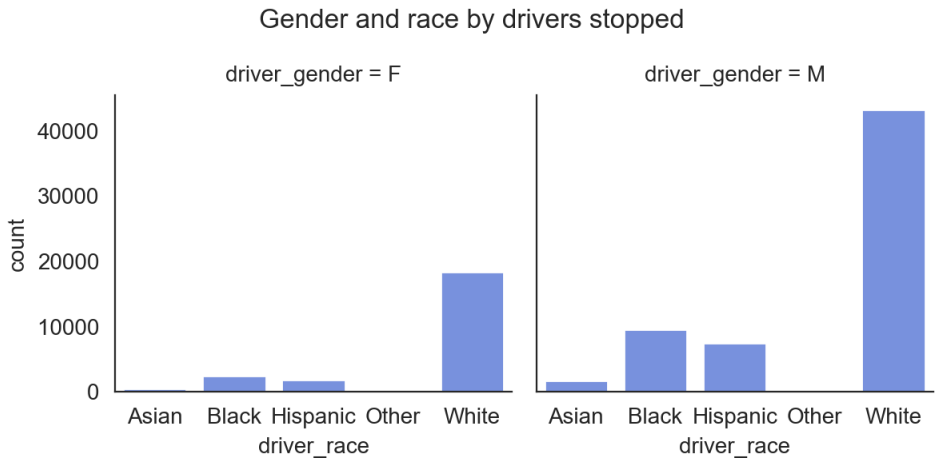


Relative Numbers
(in relation to the total number of stops)

Pay attention to the difference between absolute and relative numbers!

Possible Exemplary Solutions

- Option 1: Plots



- Option 2: Tabular

driver_gender	driver_race	Asian	Black	Hispanic	Other	White
F		0.60%	2.99%	2.17%	0.03%	21.47%
M		2.02%	11.17%	8.83%	0.25%	50.47%

1. **Always add an explanation!**
   White males are stopped most often, followed by white females, and black people

Remarks

- Again, **selection, grouping** and **aggregation** are used in the key questions respectively
- Something known from databases:

```
SELECT count(incident_number)
  FROM crimes
 GROUP BY district
```

- You are supposed to visualize and explain the results

Jupyter Notebook

- Recommended way to get started
  - Download the starter **Notebook in Moodle**
    - Exercise-3-primer.ipynp
  - Solve the exercise

- We will discuss the primers, next

Part 2: Pre-Processing (Agent)

Pre-Processing Agent

- Your task is to perform the steps of a pre-processing agent
- You first need to find a messy dataset from the internet in csv format
  - Kaggle is a great source of inspiration (i.e. Titanic Dataset)

- You are given one Jupyter-Notebooks that guides the general flow
  - agents_primer.ipynb

Data Engineering

- Garbage In – Garbage Out : Bad input data leads to bad results
- **AI models** depend on the quality and diversity of training data
- Pre-Processing is an important step, to provide cleaner data

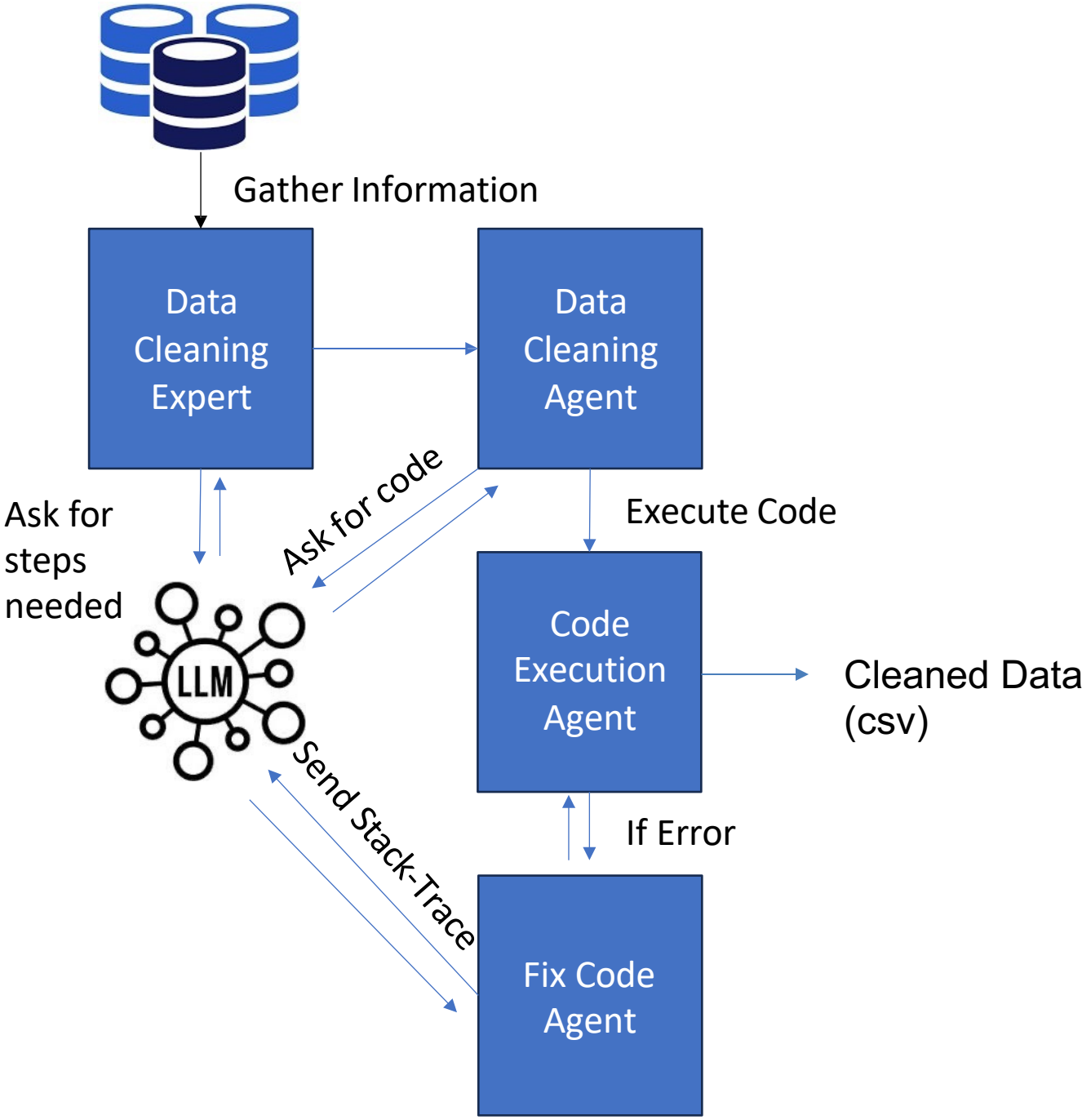
Pre-Processing – Traditional Approaches

- **Manual Effort:** Cleaning data requires human effort to identify errors, duplicates, or inconsistencies, which is tedious and prone to oversight
- **Inconsistencies:** Data coming from various sources, each using different formats, leading to inconsistencies and mismatches
- **Time-Intensive:** The larger the dataset, the more time is required to review and clean the data
- **High Error Rates:** Manual cleaning can lead to accidental data loss or inaccuracies, reducing the quality of the final dataset

Agent

- Use a pipeline where each task is handled by an AI agent

- Automates pre-processing
- 4 components
  - Data Cleaning Expert Agent
  - Data Cleaning Agent
  - Code Execution (Agent)
  - Fix Code Agent

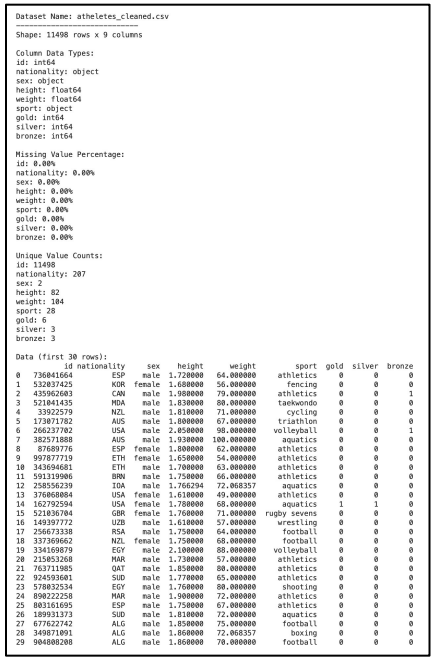


Multiple Agents

1. Data Cleaning Expert Agent
   - **Collects dataset characteristics**
   - Prompts LLM for **which steps to perform** based on the data characteristics
2. Data Cleaning Agent
   - Prompts LLM for **code to execute** the required cleaning steps
3. Code Execution Agent
   - **Executes the code** provided by the LLM
   - **We will do this manually**
4. Fix Code Agent
   - In case of an error, provide stack-trace to LLM and re-run code

Data Cleaning Expert

1. Data Cleaning Expert Agent
   - **Collects dataset characteristics**
   - **Prompts** LLM for **which steps to perform** based on the data characteristics



```
Dataset Name: atheletes_cleaned.csv
----------------------------------------
Shape: 11498 rows x 9 columns

Column Data Types:
id: int64
nationality: object
sex: object
height: float64
weight: float64
sport: object
gold: int64
silver: int64
bronze: int64

Missing Value Percentage:
id: 0.00%
nationality: 0.00%
sex: 0.00%
height: 0.00%
weight: 0.00%
sport: 0.00%
gold: 0.00%
silver: 0.00%
bronze: 0.00%

Unique Value Counts:
id: 11498
nationality: 207
sex: 2
height: 82
weight: 104
sport: 28
gold: 6
silver: 3
bronze: 3

Data (first 30 rows):
          id nationality     sex    height      weight        sport  gold  silver  bronze
0  736041664         ESP    male  1.720000   64.000000    athletics     0       0       0
1  532037425         KOR  female  1.680000   56.000000      fencing     0       0       0
2  435962603         CAN    male  1.980000   79.000000    athletics     0       0       1
3  521041435         MDA    male  1.830000   80.000000    taekwondo     0       0       0
4   33922579         NZL    male  1.810000   71.000000      cycling     0       0       0
5  173071782         AUS    male  1.800000   67.000000    triathlon     0       0       0
6  266237702         USA    male  2.050000   98.000000   volleyball     0       0       1
7  382571888         AUS    male  1.930000  100.000000      aquatics     0       0       0
8   87689776         ESP  female  1.800000   62.000000    athletics     0       0       0
9  997877719         ETH  female  1.650000   54.000000    athletics     0       0       0
10 343694681         ETH    male  1.700000   63.000000    athletics     0       0       0
11 591319906         BRN    male  1.750000   66.000000    athletics     0       0       0
12 258556239         IOA    male  1.766294   72.068357      aquatics     0       0       0
13 376068064         USA  female  1.610000   49.000000    athletics     0       0       0
14 162792594         USA  female  1.780000   68.000000      aquatics     1       1       0
15 521036704         GBR  female  1.760000   71.000000  rugby sevens     0       0       0
16 149397772         UZB    male  1.610000   57.000000     wrestling     0       0       0
17 256673338         RSA    male  1.750000   64.000000      football     0       0       0
18 337369662         NZL  female  1.750000   68.000000      football     0       0       0
19 334169879         EGY    male  2.100000   88.000000   volleyball     0       0       0
20 215053268         MAR    male  1.730000   57.000000    athletics     0       0       0
21 763711985         QAT    male  1.850000   80.000000    athletics     0       0       0
22 924593681         SUD    male  1.770000   65.000000    athletics     0       0       0
23 578032534         EGY    male  1.760000   80.000000     shooting     0       0       0
24 890222258         MAR    male  1.900000   72.000000    athletics     0       0       0
25 803161695         ESP    male  1.750000   67.000000    athletics     0       0       0
26 189931373         SUD    male  1.810000   72.000000      aquatics     0       0       0
27 677622742         ALG    male  1.850000   75.000000      football     0       0       0
28 349871091         ALG    male  1.860000   72.068357        boxing     0       0       0
29 904808208         ALG    male  1.860000   70.000000      football     0       0       0
```

Statistics

```
You are a Data Cleaning Expert. Given the following information about the data,
recommend a series of numbered steps to take to clean and preprocess it.
The steps should be tailored to the data characteristics and should be helpful
for a data cleaning agent that will be implemented.

General Steps:
Things that should be considered in the data cleaning steps:

* Removing columns if more than 40 percent of the data is missing
* Imputing missing values with the mean of the column if the column is numeric
* Imputing missing values with the mode of the column if the column is categorical
* Converting columns to the correct data type
* Removing duplicate rows
* Removing rows with missing values
* Removing rows with extreme outliers (3X the interquartile range)

Custom Steps:
* Analyze the data to determine if any additional data cleaning steps are needed.
* Recommend steps that are specific to the data provided. Include why these steps are necessary or beneficial.
* If no additional steps are needed, simply state that no additional steps are required.

IMPORTANT:
Make sure to take into account any additional user instructions that may add, remove or modify some of these steps. Include comments in your
code to explain your reasoning for each step. Include comments if something is not done because a user requested. Include comments if someth
ing is done because a user requested.

Below are summaries of all datasets provided:

Dataset Name: atheletes_cleaned.csv
----------------------------------------
Shape: 11498 rows x 9 columns

Column Data Types:
id: int64
nationality: object
sex: object
height: float64
weight: float64
sport: object
gold: int64
silver: int64
bronze: int64

Missing Value Percentage:
id: 0.00%
nationality: 0.00%
sex: 0.00%
height: 0.00%
weight: 0.00%
sport: 0.00%
gold: 0.00%
silver: 0.00%
bronze: 0.00%

Unique Value Counts:
id: 11498
nationality: 207
sex: 2
height: 82
weight: 104
sport: 28
gold: 6
silver: 3
bronze: 3
```

Prompt

LLM – Recommended Steps

- 1. Load and Inspect the Dataset
- 2. Remove Columns With More Than 40 Percent Missing
- 3. Convert Columns to Appropriate Data Types
- 4. Impute Missing Numeric Values (height and weight) Us
- 5. Remove Duplicate Rows
- 6. Remove Rows With Remaining Missing Values
- 7. Detect and Remove Extreme Outliers Using the 3x IQR
- 8. Validate Categorical Consistency
- 9. Analyze the Data for Additional Cleaning Needs
- 9b. Ensure medal counts are non-negative
- 10. Confirm Data Types Again After Cleanin

Numbered Data Cleaning Steps

1. Load and Inspect the Dataset

python — Code kopieren

```
df = pd.read_csv("athletes.csv")
# Inspect shape, dtypes, missing values, unique values to verify metadata
```

This confirms that the file matches the provided summaries before transformation begins.

2. Remove Columns With More Than 40 Percent Missing Data

python — Code kopieren

```
# None of the columns exceed 40 percent missing based on metadata
# Comment to explain: No columns removed because none surpass the threshold.
```

No columns qualify for removal since height has about 2.86 percent missing and weight about 5.71 percent missing.

3. Convert Columns to Appropriate Data Types

python — Code kopieren

```
# Convert medal count columns to integer
df["gold"] = df["gold"].astype(int)
df["silver"] = df["silver"].astype(int)
df["bronze"] = df["bronze"].astype(int)
# Confirm id stays int and categorical columns stay object
```

These medal columns are numeric though stored as object, so converting improves consistency.

4. Impute Missing Numeric Values (height and weight) Using Mean

python — Code kopieren

```
df["height"].fillna(df["height"].mean(), inplace=True)
df["weight"].fillna(df["weight"].mean(), inplace=True)
```

This aligns with the general rules since both columns are numeric.

5. Remove Duplicate Rows

python — Code kopieren

```
df.drop_duplicates(inplace=True)
```

The id column has 11538 unique values out of 11543 rows, so some duplicates likely exist. Deduplication ensures row uniqueness.

6. Remove Rows With Remaining Missing Values

Data Cleaning Agent

- **Prompts LLM** for code to execute the required cleaning steps

```
You are a Data Cleaning Agent. Your job is to create a clean_data() function that can be run on the data provided using the following recomm
ended steps.

Recommended Steps:
1. Load and Inspect the Dataset
2. Remove Columns With More Than 40 Percent Missing Data
3. Convert Columns to Appropriate Data Types
4. Impute Missing Numeric Values (height and weight) Using Mean
5. Remove Duplicate Rows
6. Remove Rows With Remaining Missing Values
7. Detect and Remove Extreme Outliers Using the 3x IQR Rule
8. Validate Categorical Consistency
9. Analyze the Data for Additional Cleaning Needs
9b. Ensure medal counts are non-negative
10. Confirm Data Types Again After Cleaning

You can use Pandas, Numpy, and Scikit Learn libraries to clean the data.

Below are summaries of all datasets provided. Use this information about the data to help determine how to clean the data:


Dataset Name: athletes.csv
------------------------------
Shape: 11543 rows x 9 columns

Column Data Types:
id: int64
nationality: object
sex: object
height: float64
weight: float64
sport: object
gold: object
silver: object
bronze: object

Missing Value Percentage:
weight: 5.71%
height: 2.86%
id: 0.00%
nationality: 0.00%
sex: 0.00%
sport: 0.00%
gold: 0.00%
silver: 0.00%
bronze: 0.00%

Unique Value Counts:
id: 11538
nationality: 207
sex: 2
height: 82
weight: 123
sport: 28
gold: 7
silver: 4
bronze: 4

Data (first 30 rows):
          id nationality     sex  height  weight         sport gold silver bronze
0  736041664         ESP    male    1.72    64.0     athletics    0      0      0
1  532037425         KOR  female    1.68    56.0       fencing    0      0      0
2  435962603         CAN    male    1.98    79.0     athletics    0      0      1
3  521041435         MDA    male    1.83    80.0    taekwondo    0      0      0
```

Prompt

Code Execution Agent

- Runs code locally
- In case there are errors, run fix code agent

- Finally, check, if the data is successfully cleaned by starting the workflow again

Nachgedacht für 43s ›

python — Code kopieren

```python
def clean_data(data_raw):
    import pandas as pd
    import numpy as np
    from sklearn.impute import SimpleImputer

    # Make a copy to avoid mutating the original input
    df = data_raw.copy()

    # 1. "Load and Inspect the Dataset"
    # In this function, the dataset is already loaded and passed in as data_raw.
    # Here we conceptually "inspect" by checking basic structure if needed (no prints t
    # Example (optional for debugging only):
    # print(df.head())
    # print(df.info())

    # 2. Remove Columns With More Than 40 Percent Missing Data
    # Based on the metadata, none of the columns should be removed,
    # but we keep this logic generic so it will work even if the data changes.
    missing_fraction = df.isna().mean()
    cols_to_drop = missing_fraction[missing_fraction > 0.40].index
    # Comment: No columns are expected to be dropped here for athletes.csv,
    # but the code supports the general rule.
    df = df.drop(columns=list(cols_to_drop), errors="ignore")

    # 3. Convert Columns to Appropriate Data Types
    # Ensure "id" is numeric (int). If it is already int64, this is a no-op.
    if "id" in df.columns:
        df["id"] = pd.to_numeric(df["id"], errors="coerce")
        # Any non-numeric ids become NaN and will be handled in the missing data step.

    # Medal columns are stored as object but represent integer counts.
    # Convert them to numeric, coercing invalid entries to NaN for later cleaning.
    for col in ["gold", "silver", "bronze"]:
        if col in df.columns:
            df[col] = pd.to_numeric(df[col], errors="coerce")

    # 4. Impute Missing Numeric Values (height and weight) Using Mean
    # We follow the recommendation to use the mean for numeric imputation.
    # Use SimpleImputer from scikit-learn and ravel the result as required.
    imputer = SimpleImputer(strategy="mean")

    for col in ["height", "weight"]:
        if col in df.columns:
            # fit_transform returns a 2D array, but a Series column is 1D,
            # so we ravel the array before assignment (best practice).
            df[col] = imputer.fit_transform(df[[col]]).ravel()

    # 5. Remove Duplicate Rows
    # Drops fully duplicated rows to avoid repeated records.
    df = df.drop_duplicates()

    # 6. Remove Rows With Remaining Missing Values
```

Submission Deadline: Tue, 06.01.26

- The exercise is considered to have been **successfully solved,** if
  - Part 1: An answer backed by evidence has been given for each of the **6 key questions**
  - Part 2: Pre-processing workflow **was successfully executed and data is cleaned**
- **Deliverables:**
  - **Part 1: EDA (2 Files)**
    a) **Executed** notebook
    b) A **html export** of the **executed notebook**
  - **Part 2: Agent** (**6 Files**):
    a) Executed Notebook and HTML export
    b) Original and cleaned CSV file
    c) Summary Statistics before and after cleaning

Bonusblatt

Questions?